# Disaster Alert & Resource Allocation System

System Analysis & Design Case Study | Menofia University

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Faculty of Computers and Information

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| **Project Title** | Disaster Alert & Resource Allocation System |
| **Project Team** | Akram EL Hewehy, Mahmoud Fetouh, Yousef Hafez, Yousef EL Maasarawy, Ahmed Fayed (Section 2) |
| **Methodological Reference** | System Analysis Case Study - Data Flow Diagram (Section 4 Standard) |

## 1. Functional Requirements for Disaster Alert System

Based on the functional requirements format presented in Section 4 slide examples:

- **Admin** should be able to add, update, and delete disaster alerts.
- **Admin** should be able to add, update, and delete system users (field teams and citizens).
- **Admin** should be able to manage resource allocation schedules.
- **Field Agencies / Instructors** should be able to view allocation tasks and update resource availability status.
- **Citizens / Students** should be able to send emergency help requests and view shelter locations.
- **Citizens / Students** should be able to receive disaster alerts and view emergency status reports.

## 2. Identification of DFD Elements

**External Entities (Double Square - Noun):** Admin, Citizen, Field Agency.

**Processes (Rounded Rectangle - Verb-Adjective-Noun):** 1.0 Manage Disaster Alerts, 2.0 Manage Users, 3.0 Process Emergency Request, 4.0 Show Results / Status Reports, 5.0 Allocate & Dispatch Resources.

**Data Stores (Open-Ended Rectangle - Noun with ID):** D1 Disaster Alert, D2 Citizens, D3 Field Agencies, D4 Emergency Request, D5 Dispatch Log, D6 Resources.

**Data Flows (Arrow - Noun):** Alert data, User data, Emergency request, Dispatch order, Status report, Confirmation.

## 3. Context Level DFD (Level 0 Overview)

As per Section 4 context model, Process 0 represents the entire system with zero data stores:

**Process 0:** Disaster Alert & Resource Allocation System

- **Admin Inputs/Outputs:** Sends *Alert data*, *Users data*, and *Schedule* to System (0); receives *Confirmation*.
- **Field Agency Inputs/Outputs:** Receives *Dispatch Order* from System (0); sends *Resource Status Update*.
- **Citizen Inputs/Outputs:** Sends *Emergency Request* to System (0); receives *Alert Notification* and *Status Report*.

## 4. Diagram 0 (Level 0 DFD Explosion)

Explosion of Process 0 into main system processes (numbered 1.0 to 5.0) and major data stores:

| Process # | Process Name | Data Flow & Data Store Connections |
|---|---|---|
| **1.0** | **Manage Disaster Alerts** | Receives *Alert data* from Admin → Sends *Updated Alert data* to **D1 Disaster Alert** → Returns *Confirmation* to Admin. |
| **2.0** | **Manage Users** | Receives *Users data* from Admin → Updates **D2 Citizens** and **D3 Field Agencies** → Returns *Confirmation*. |
| **3.0** | **Process Emergency Request** | Receives *Emergency Request* from Citizen → Reads/Writes *Request Details* to **D4 Emergency Request**. |
| **4.0** | **Show Results** | Reads *Report data* from **D5 Dispatch Log** → Displays *Status Report / Results* to Citizen. |
| **5.0** | **Allocate & Dispatch Resources** | Exchanges *Resource allocation* with Field Agency and updates **D6 Resources**. |

## 5. Child Diagram (Explosion of Process 3.0)

Exploding Process 3.0 (Process Emergency Request) into child processes following Section 4 numbering standard (3.1, 3.2):

- **Process 3.1 (Select Distress Request):** Receives *Emergency Request* from Citizen → Stores *Selected Request* into **D4 Emergency Request**.
- **Process 3.2 (Submit Request Details):** Reads *Request Details* from **D4 Emergency Request** → Sends *Validated Request* to **D5 Dispatch Log** and returns *Confirmation* to Citizen.

## 6. Section 4 DFD Rules & Error Check

| Section 4 DFD Rule | Compliance Verification |
|---|---|
| **No Direct Entity-to-Entity Flow** | All data flows pass through a process. No direct entity-to-entity or entity-to-datastore arrows exist. |
| **No Miracle or Black Hole** | Every process has at least one input and at least one output data flow. |
| **Naming Conventions** | Processes are labeled using `Verb-Adjective-Noun`; Entities and Data Stores use `Nouns`. |
| **Balanced Decomposition** | Input/output data flows of parent Process 3.0 match the interface flows of Child Diagram 3.0. |